S1 Appendix
Coevolutionary balance of resting-state brain networks in autism

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This supplement accompanies the corrected revision analysis. The main Methods distinguish downloaded PCP fALFF derivatives from local regional aggregation and describe the limits of upstream documentation. S2 Data contains the full machine-readable results; tables here summarize selected checks.

Cohort, coverage, and cortical assignment

The original requested pool has 182 participants (90 ASD/92 controls). Age 18–30 and observed FIQ > 80 yield 140 (69/71). This revision does not reconstruct a randomly sampled 100/100 cohort. All available participants were recorded male; handedness is not an exclusion. Technical exclusions diagnosed with all 200 parcels in Stage 1 are not automatically carried into the primary 106-node design; all primary participants must instead satisfy the frozen selected-node QC rule.

The primary model omits CMU ($n = 1$ ASD), SBL ($n = 1$ ASD), and UM_2 ($n = 2$ controls) under the site-size rule, leaving 136 (67/69). Broader-181 and full-200 feature cohorts contain 104 (48/56) and 69 (31/38), and their adjusted models contain 101 (47/54) and 64 (30/34). Full per-design and trimming manifests are included in S2 Data.

Of the primary 106 nodes, 13 are visual, 16 somatomotor, nine dorsal attention, 24 salience/ventral attention, one limbic, 14 frontoparietal, 14 default, and 15 unassigned. Global quantities retain all 106; cortical block sums omit edges involving unassigned nodes. A within-limbic pair does not exist. Three within-limbic quantities and the legacy bipolarity descriptor are unavailable or excluded from the 88-feature inferential schema. The same schema, including both graph efficiencies, is used in both pipelines for clinical and predictive analyses. The complete ROI list and assignment table are provided in S2.

Table S1 gives the primary model site-by-diagnosis counts.

Table S1: Primary adjusted-model site counts. CALTECH contributes only ASD participants.

SITE_ID	ASD	Control
CALTECH	9	0
LEUVEN_1	11	5
MAX_MUN	5	12
NYU	12	19
OLIN	4	3
PITT	4	8
TRINITY	7	7
UM_1	1	2
USM	14	13

Site diagnostics and restricted sensitivity

For each design and pipeline, the original adjusted-model participant list is joined to saved raw and ComBat feature exports. The model matrix contains an intercept, diagnosis, standardized age, FIQ, FD, and site

indicators. All designs have full rank. Site partial R^2 is $(\mathrm{SSE}_{\text{without site}} - \mathrm{SSE}_{\text{with site}})/\mathrm{SSE}_{\text{without site}}$. It describes the additional fit attributable to site under this model. Median-centered Levene statistics compare biological- covariate residual dispersion across sites. These post-ComBat summaries are exploratory descriptive diagnostics on the fitting sample, not independent tests that harmonization succeeded. No diagnostic flag triggers participant removal. The main manuscript now retains dedicated within-network and between-network Results sections and full cortical effect figures. Gray cells are structurally unavailable; no primary cortical test survives its family correction. the supporting figures below displays the exact residualized global-efficiency comparisons underlying the saved partial correlations.

Raw primary maximum Cook's distance is .080 with GSR and .062 without GSR; maximum leverage is .352. HC3 uncertainty accounts for leverage but does not supply site-clustered inference. Influence records, site/group H summaries, and all diagnostic values are retained in S2 Data.

The mixed-diagnosis-site restriction excludes CALTECH's nine ASD participants, leaving 127 participants (58 ASD and 69 controls) at eight sites. Adjusted coefficients are 1.115 (95% CI −17.531, 19.762) with GSR and 1.660 (−12.775, 16.096) without GSR; both Holm $p = 1$. Larger-node models already contain only mixed-diagnosis sites and therefore retain their original estimates. This is a raw adjusted sensitivity; we do not refit ComBat in the restricted cohort.

Table S2 summarizes the primary site and influence diagnostics.

Table S2: Primary-model diagnostics. Site partial R^2 is descriptive and conditional on diagnosis, age, FIQ and mean FD. The design is identical before and after ComBat.

Pipeline	Version	Site partial R2	Max leverage	Max Cook's D
GSR	raw	0.056	0.352	0.080
GSR	ComBat	0.004	0.352	0.079
noGSR	raw	0.047	0.352	0.062
noGSR	ComBat	0.005	0.352	0.072

Node-set sensitivities

the supporting figures below shows raw adjusted Hamiltonian coefficients. Each design has its own participant set and total edge count. Raw energy magnitudes are therefore not interchangeable across designs. Complete estimates for ComBat and pooled trimming, including participant-specific exclusions and trimming fences, are in S2 Data. No corrected diagnosis effect is detected in these models; sensitivity agreement does not demonstrate equivalence.

Original-analysis continuity and H–metric complements

The original manuscript distinguishes global, within-network, between-network, within-group H relationships, clinical Pearson/Spearman correlations, PLSR, PLSC, and classification, with GSR/no-GSR sensitivity. These questions are now again named explicitly in the main Results. The complete original-to-revision map, significant association lists, nominal-only group results, and original export reproduction checks are included under continuity/ in S2.

All 364 original exported H–feature Pearson correlations, raw probabilities and per-group FDR probabilities were reproduced numerically from the supplied original feature tables to absolute tolerance 10^{-10}. The original no-GSR ASD H–global-efficiency association is $r = -.324338$, $p = .003334$, $q = .005917$ ($n = 80$). It is therefore present in those original data. That source sample and feature construction differ from the revised common-node analysis; replication of an export is not confirmation of the exact submitted run or proof of a group interaction. The revised raw-eligible correlations use 69 ASD and 71 controls, whereas matched raw/ComBat complements use 67 and 69.

Table S3 reports both matched versions, including nonsignificant blocks. The 348-test alternative pools diagnosis/pipeline blocks; it can change a subgroup's q-values in either direction and is not described as uniformly stricter. Shared Hamiltonian/block terms and proportional weighted-FC/energy features mean

significant counts are not independent biological discoveries. The original negative efficiency association is not recovered in the revised raw eligible feature definition, independently of nuisance adjustment.

Table S3: Descriptive Pearson H associations on identical model participants before/after ComBat. q87 uses per-group correction; q348 pools all four pipeline/group blocks per version. Neither correction scope is uniformly more conservative for every subgroup.

Version	Pipeline	Group	N	q87<.05	q348<.05
ComBat matched	GSR	ASD	67	24	22
ComBat matched	GSR	Control	69	25	18
ComBat matched	noGSR	ASD	67	10	10
ComBat matched	noGSR	Control	69	0	10
Raw matched	GSR	ASD	67	28	18
Raw matched	GSR	Control	69	19	19
Raw matched	noGSR	ASD	67	10	12
Raw matched	noGSR	Control	69	0	6

Table S4 reports the original spectral descriptor as a separate exploratory raw adjusted sensitivity. Its two tests are Holm corrected separately from the frozen primary/secondary families; it remains absent from clinical and prediction feature schemas. Both adjusted probabilities are .916.

Table S4: Original spectral bipolarity descriptor: additional exploratory raw adjusted diagnosis models. Holm covers the two pipeline tests, separately from the frozen inferential families. The descriptor is not asserted to optimize signed structural balance.

Pipeline	N	Coefficient	CI low	CI high	Holm p
GSR	136	0.0004	-0.0067	0.0075	0.916
noGSR	136	0.0105	-0.0175	0.0386	0.916

Table S5 retains the original null-model question in both diagnostic groups. These individual tests use a pooled 560-comparison exploratory BH family; the separate 71-control baseline retains its explicit 284-comparison correction. Neither family tests an ASD–control difference in H or null z.

Table S5: Separate diagnostic-group summaries of the existing state-permutation nulls. Exploratory BH covers all 560 participant/pipeline/null comparisons here. Control-only 284-test correction is a different explicitly labeled family.

Group	Pipeline	Null	N	Mean z	q<.05 count
ASD	GSR	Blocked	69	-2.331	40
ASD	GSR	Unrestricted	69	-4.705	61
ASD	noGSR	Blocked	69	-2.591	46
ASD	noGSR	Unrestricted	69	-4.917	60
Control	GSR	Blocked	71	-2.313	36
Control	GSR	Unrestricted	71	-4.690	61
Control	noGSR	Blocked	71	-2.760	50
Control	noGSR	Unrestricted	71	-5.159	64

Clinical analyses

The five scores are ADI-R social ($n = 30$), ADI-R verbal ($n = 30$), ADI-R onset ($n = 28$), ADOS total ($n = 50$), and ADOS communication ($n = 43$), identically across pipelines. Negative/sentinel values are missing. Feature correlations use all available 88 features. Adjusted Hamiltonian correlations use age, IQ, mean FD, and site. No feature or adjusted Hamiltonian correlation survives its specified BH correction.

The ADOS and five-score complete-case PLSC blocks are exploratory; rank-wise permutation probabilities across all reported latent variables are BH corrected. A nominal no-GSR five-score LV3 $p = .00699$ becomes $q = .0979$. All PLSC and correlation rows, including nonsignificant tests, are included in S2. the supporting figures below displays all adjusted H correlations and OOF PLSR scores. The prior loading figures came from superseded fitted models; they are not reused as current component estimates.

Table S6 lists every score-specific PLSR validation result.

Table S6: Nested site-held-out PLS regression. Baselines predict the outer-fold training mean. Negative OOF R^2 does not imply evidence of an inverse biological effect.

Pipeline	Score	N	OOF R2	OOF MSE	Baseline MSE
GSR	ADI R Social	30	-0.416	37.895	27.874
GSR	ADI R Verbal	30	-0.130	25.895	23.572
GSR	ADI R Onset	28	-0.928	3.512	2.017
GSR	ADOS Total	50	-0.029	10.838	10.723
GSR	ADOS Communication	43	-0.047	1.409	1.392
noGSR	ADI R Social	30	-0.381	36.939	27.874
noGSR	ADI R Verbal	30	-0.370	31.387	23.572
noGSR	ADI R Onset	28	-0.348	2.455	2.017
noGSR	ADOS Total	50	-0.105	11.633	10.723
noGSR	ADOS Communication	43	-0.305	1.756	1.392

Male-control null baseline reported separately

All 71 eligible male controls from nine sites are included in this baseline, including controls at sites too small for the group regression. No energy-based trimming is applied. For each of the two pipelines and two state-permutation schemes there are exactly 71 distinct control records (284 comparisons total). Each record retains the original 1,000-permutation Monte Carlo probability. An exploratory BH correction spans all 284 individual comparisons. The counts of lower-tail $p < .05$ are 64/69 for unrestricted GSR/no-GSR and 40/53 for blocked GSR/no-GSR; the corresponding BH $q < .05$ counts are 61/64 and 36/50. These counts do not test ASD versus controls.

Mean control z is -4.690 with GSR and -5.159 without GSR for unrestricted states, versus -2.313 and -2.760 within network/hemisphere blocks. the supporting figures below displays every control's score. Mean empirical-minus-null energy intervals resample sites 5,000 times (seed 20261026), computing the participant-weighted mean in each resampled set of whole site clusters. The percentile intervals condition on the saved null means and do not include all Monte Carlo estimation uncertainty or constitute a pooled randomization test. S2 contains the complete control-only table and all interval endpoints.

These null configurations hold each participant's full signed weighted FC matrix fixed and randomize only state assignment. They address whether observed relative fALFF states align with that FC more than randomized states do. They do not test against degree-rewired or signed-weight-shuffled networks. The two latter nulls would ask additional questions and are not claimed to have been run here.

Randomization and mathematical coupling

Unrestricted permutations preserve FC and the total number of nodal labels. Network/hemisphere permutations additionally preserve state counts within the specified blocks. Their lower-tail probabilities test alignment under those randomizations, not an autism group effect. Small blocks may offer few distinct permutations, and neither scheme preserves full spatial autocorrelation. No primary participant has zero null variance. The participant records include the number of positive states, observed energy, null mean/SD, Monte Carlo probability, and the coefficient linking a uniform weight shift to energy.

For n nodes with $s_i \in \{-1, +1\}$, $\sum_{i<j} s_i s_j = ((\sum_i s_i)^2 - n)/2$. Thus adding a constant c to each Fisher weight changes H by $[n - (\sum_i s_i)^2]c/2$. With 53 positive and 53 negative labels the coefficient is 53, rather

than the 100 appropriate to exactly balanced 200-node states. Median ties can change that coefficient and are not forced into artificial equal halves.

Table S7 summarizes both diagnosis groups together; the control-only baseline above is reported separately.

Table S7: Descriptive participant null-score summaries. These are alignment randomizations, not diagnostic tests or full spatial-autocorrelation nulls.

Pipeline	Null	N	Mean z	SD z
GSR	Network/hemisphere	140	-2.322	1.449
GSR	Unrestricted	140	-4.698	2.480
noGSR	Network/hemisphere	140	-2.677	1.631
noGSR	Unrestricted	140	-5.039	2.432

Prediction and reproducibility

Outer classification folds have 31, 28, 27, 27, and 27 participants. Sites are disjoint between train and test; both classes are present in every inner and outer split. All 10 completed classifier/pipeline pairs have predictions for all 140 participants. Hyperparameter selection is restricted to inner folds, and ComBat is not used. Feature availability and node QC are cohort-wide, limiting the analysis to exploratory prediction conditional on the fixed feature schema. OOF intervals bootstrap sites from fixed predictions; they do not repeatedly refit the complete training and QC process. the supporting figures below use all 140 OOF predictions for each model and pipeline. No best classifier is selected using those plots; pooled ROC curves remain descriptive in this site-held-out setting.

The original-algorithm extension currently covers 5 model types (logistic regression, RBF SVM, Gaussian naive Bayes, KNN, XGBoost), with 10 complete pipeline/model pairs. All five original model types are now represented. The prior logistic/SVM records are unchanged. The extension plan, dependency versions, fold checkpoints, and combined outputs are supplied in S2.

The final repair reuses the 626 feature rows with SHA-256

```
25178d5b3f5b87ad10c4b2f60f0574fb5fe51b26d042ddac7255f9030894848b
```

The initial optional power-diagnostic numerical failure was corrected without changing feature extraction. The initial incomplete OOF classifier aggregates were replaced. Original uploads and the submitted manuscript are preserved. The manifest in S2 indexes the exact supporting tables and their checksums.

Feature-selection audit and exploratory extension

The completed baseline contains all five original algorithm types. All 50 saved outer-training models were reconstructed using their saved settings without new hyperparameter search; all predicted labels match. ANOVA selection chose 25% (22 features) in 11 fits, 50% (44 features) in 17, and 100% in 22. Per-fold selected feature names and reconstruction score differences are supplied in S2. The baseline no-reduction option is part of its inner selection search, not an absence of feature selection implementation.

The follow-up's training-only redundancy pruning reduced 88 to 59 columns in every outer fit, followed by all-features, ANOVA, mutual-information or L1-ranked selection chosen jointly with classifier parameters in the inner folds. All 50 follow-up outer fits have complete predictions, with 140 unique participants for every algorithm/pipeline. Selection counts are 5/10/20 or all 59. The global and cortical group models and the 59 reported H associations are unchanged.

the supporting figures below reports all model performance estimates before/after the follow-up. Four balanced-accuracy point estimates rise and six fall; every paired change interval includes zero. The largest positive change is no-GSR logistic regression, .059 (conditional 95% interval −.0148 to .1088). The no-GSR KNN follow-up has BA=.557 and descriptive pooled AUC=.606. These estimates do not establish a generalizable improvement or justify selecting a model after examining test performance. The feature representation and compact XGBoost grid also changed, precluding attribution of a difference to selector method alone. The prespecified follow-up document explicitly records that the baseline outer results had already been viewed.

the supporting figures below give follow-up ROC curves and row-normalized confusion heatmaps with exact counts. the supporting figures below shows 25 highest mean selected-feature frequencies, while all 88 features and all fold selections are provided as CSV. Frequencies are descriptive across five overlapping training folds, not independent stability tests, biomarker significance, or permutation importance. Corresponding baseline feature-selection frequency and normalized confusion heatmaps are supplied as standalone figures. No held-out test observation is used to train a selector or choose its parameters.

Supporting figures

Figure S1 presents composition of the primary node set.

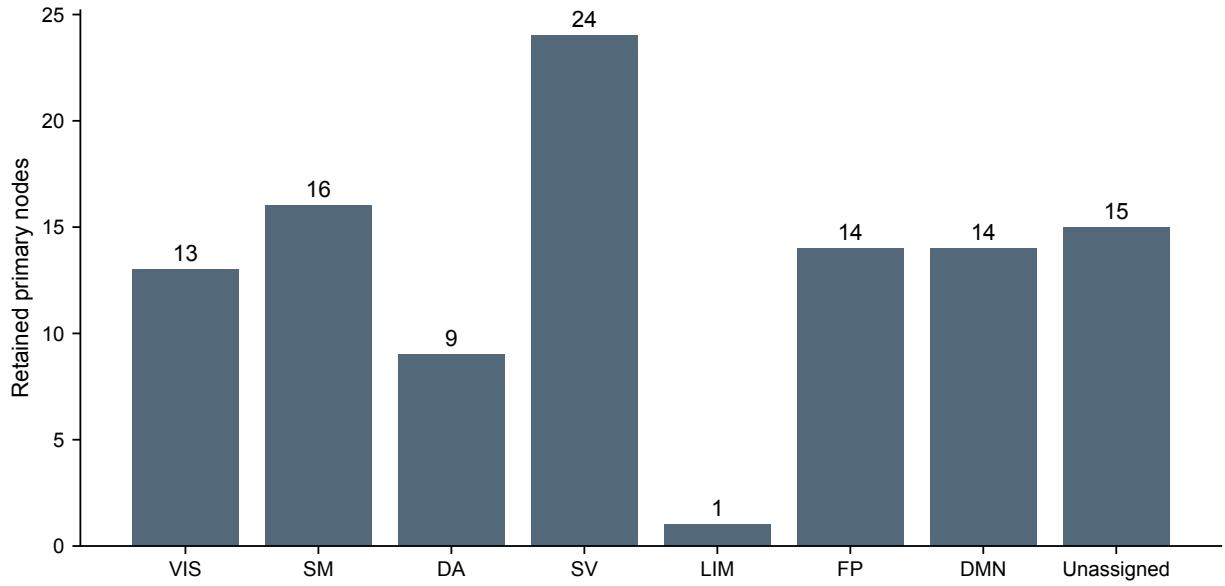


Figure S1: **Composition of the primary node set.** The common 106-node graph contains 91 cortical-assigned and 15 unassigned nodes. The seven network counts are shown. A single limbic node cannot support within-network edges.

Figure S2 presents raw h estimates across node-set designs.

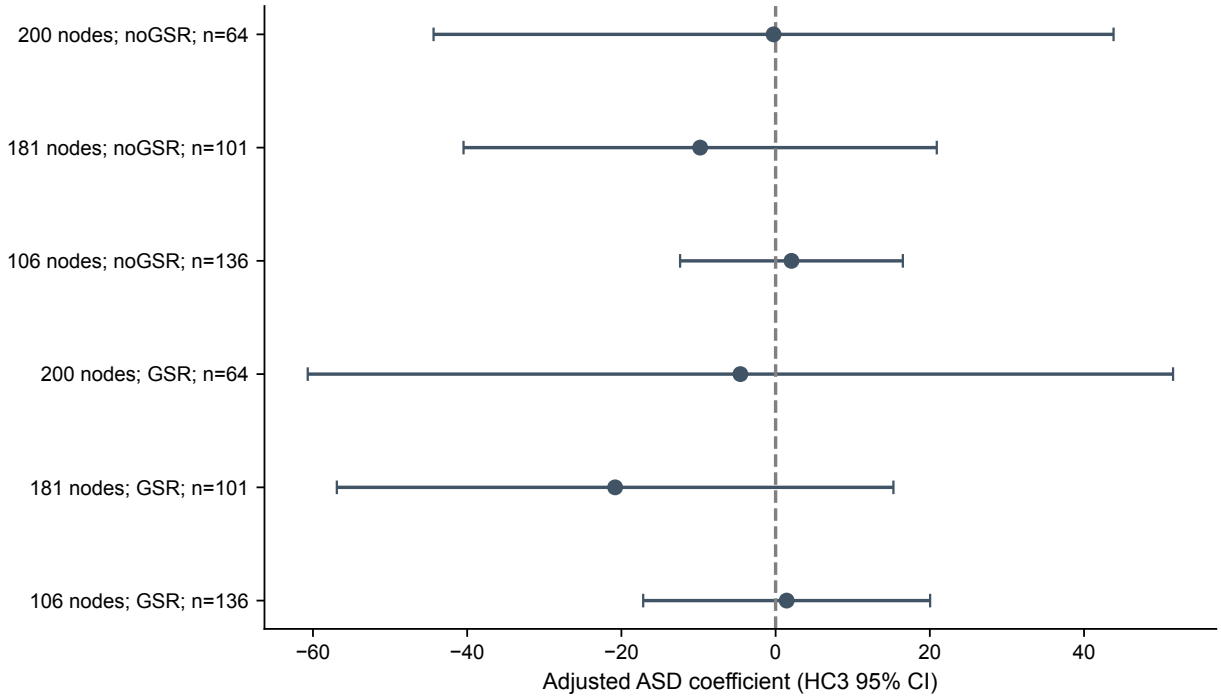


Figure S2: **Raw H estimates across node-set designs.** Points show adjusted diagnosis coefficients and HC3 95% intervals. Node and participant sets differ across designs; these comparisons cannot isolate a node-count effect.

Figure S3 presents individual male-control empirical and null energies.

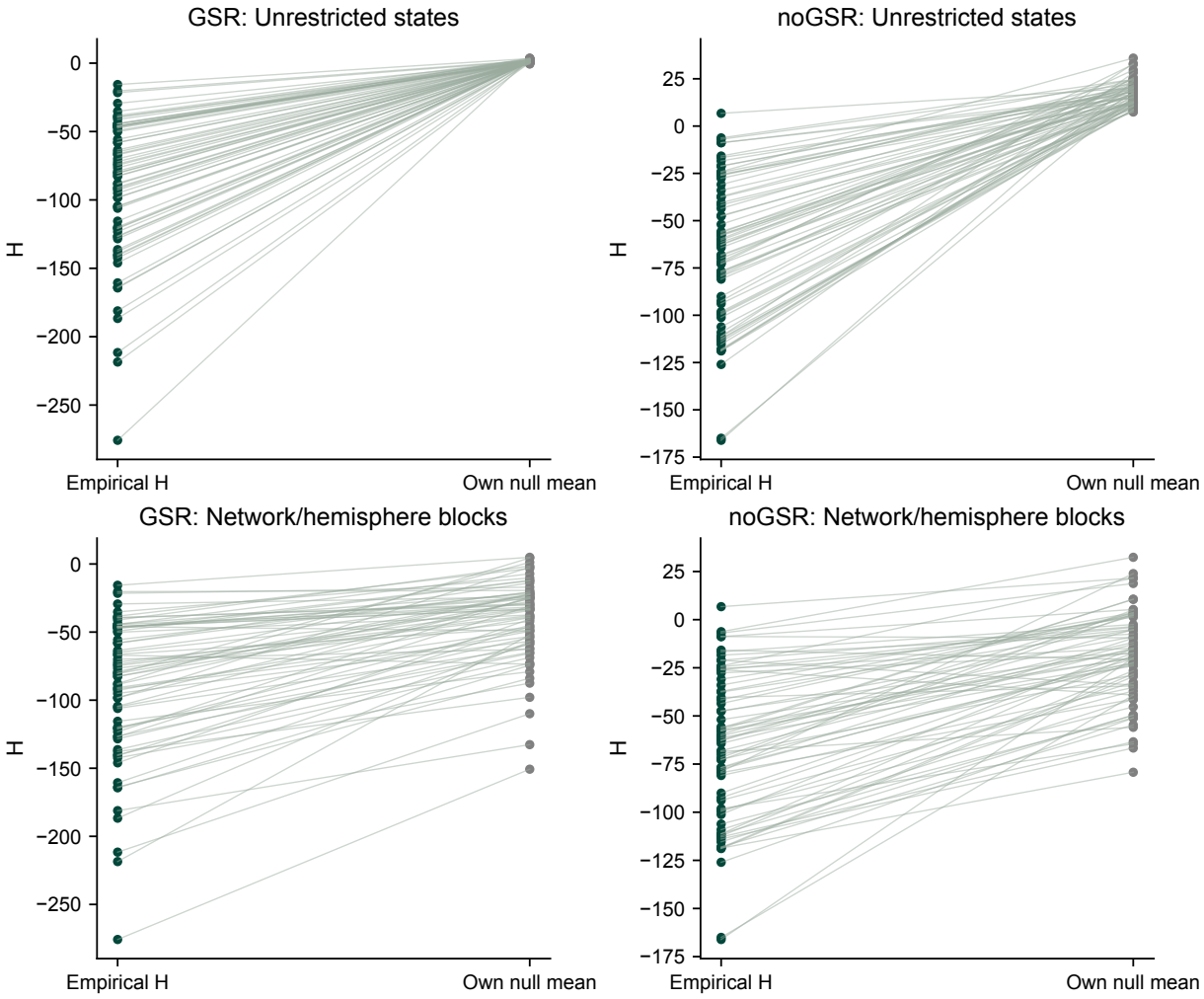


Figure S3: **Individual male-control empirical and null energies.** Each line connects the empirical H of one of 71 controls to that participant's mean of 1,000 randomizations. Pipelines and null constraints are separated. No topology rewiring is performed.

Figure S4 presents participant-level null z-scores.

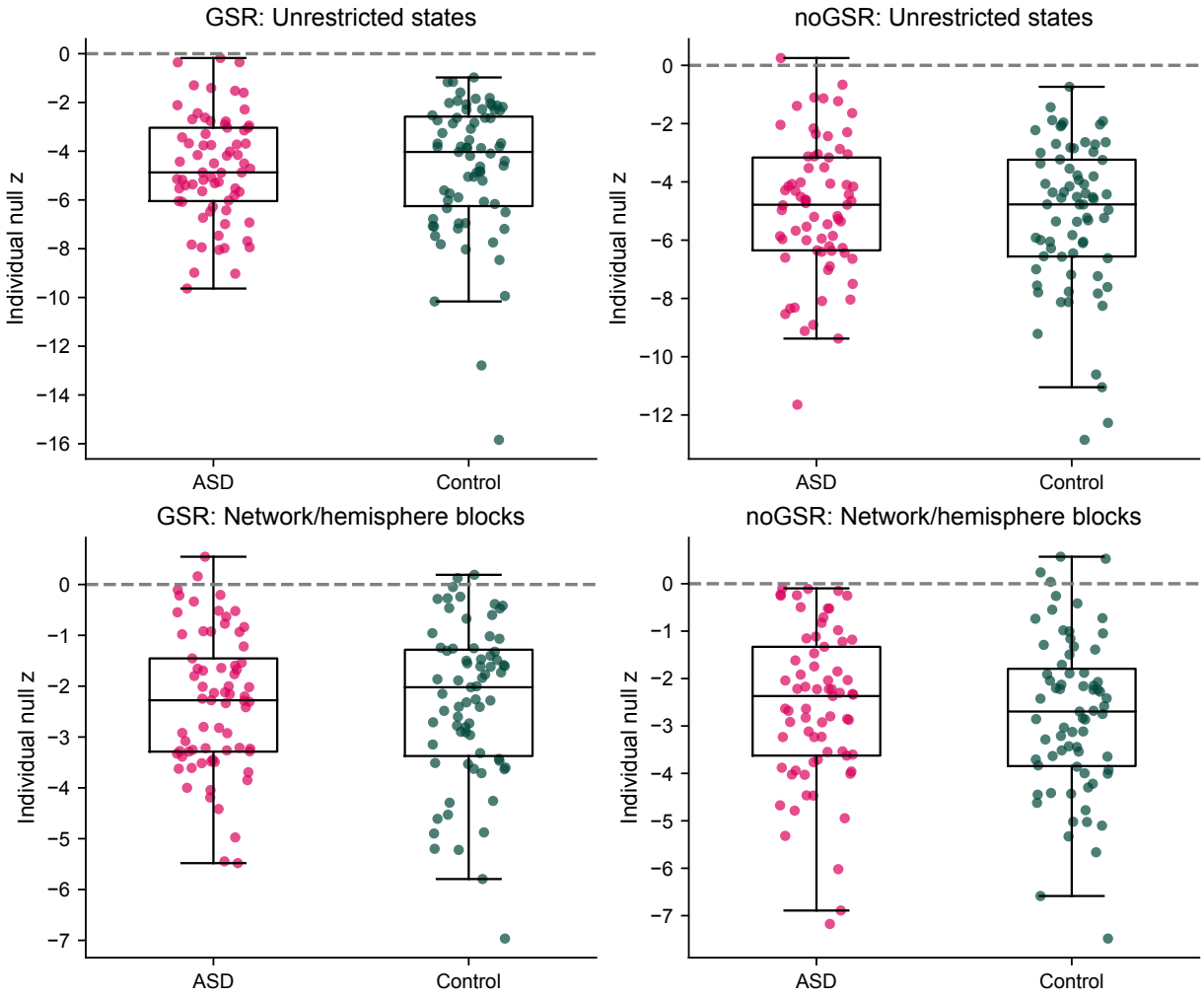


Figure S4: **Participant-level null z-scores.** Both groups are shown under unrestricted and blocked state permutations. Negative z denotes H below the individual null mean. These are alignment comparisons, not diagnosis tests.

Figure S5 presents within-network block-energy distributions.

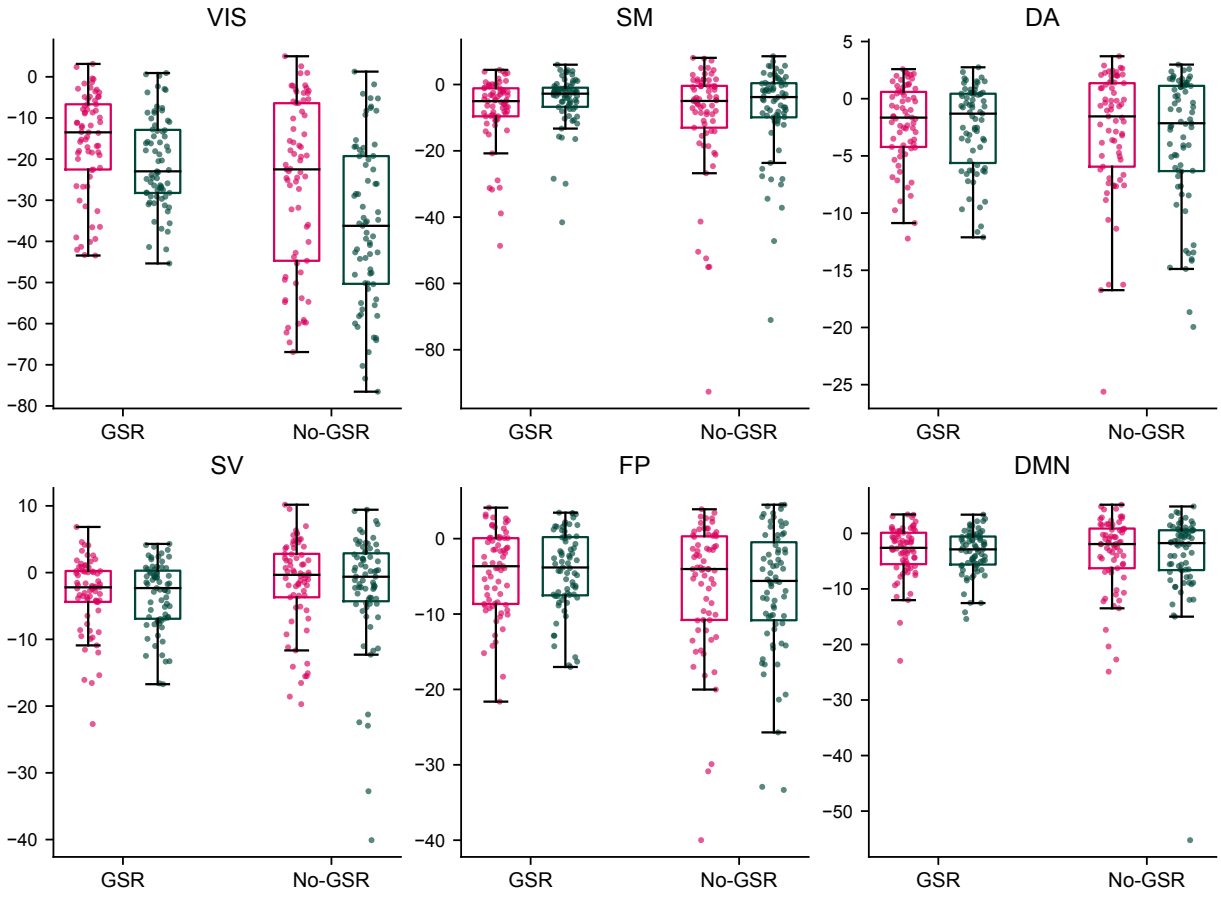


Figure S5: **Within-network block-energy distributions.** All six available within-network energy blocks are shown in both pipelines. ASD is magenta and controls dark green. Descriptive distributions use all 140 eligible participants, whereas adjusted comparisons use 136.

Figure S6 presents baseline classifier confusion matrices.

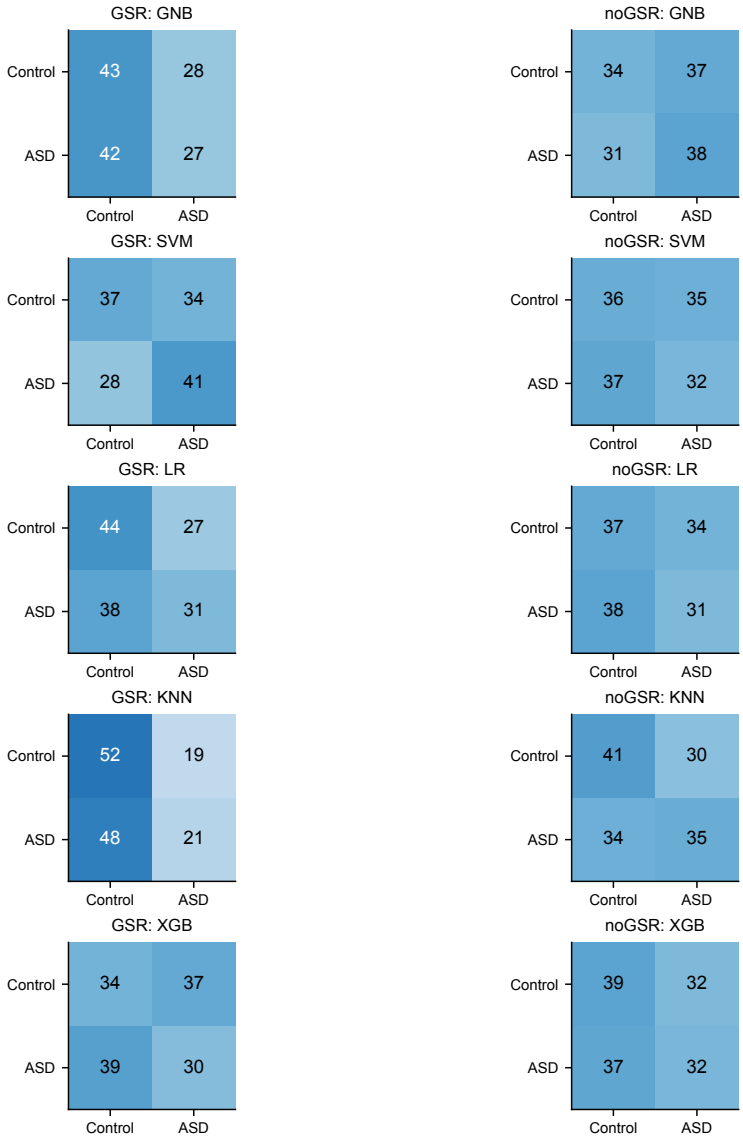


Figure S6: **Baseline classifier confusion matrices.** Each matrix contains all 140 complete out-of-fold predictions. Counts are printed and color encodes true-class-normalized proportions for every model and pipeline. No model is selected from outer-fold performance.

Figure S7 presents descriptive principal-component projection.

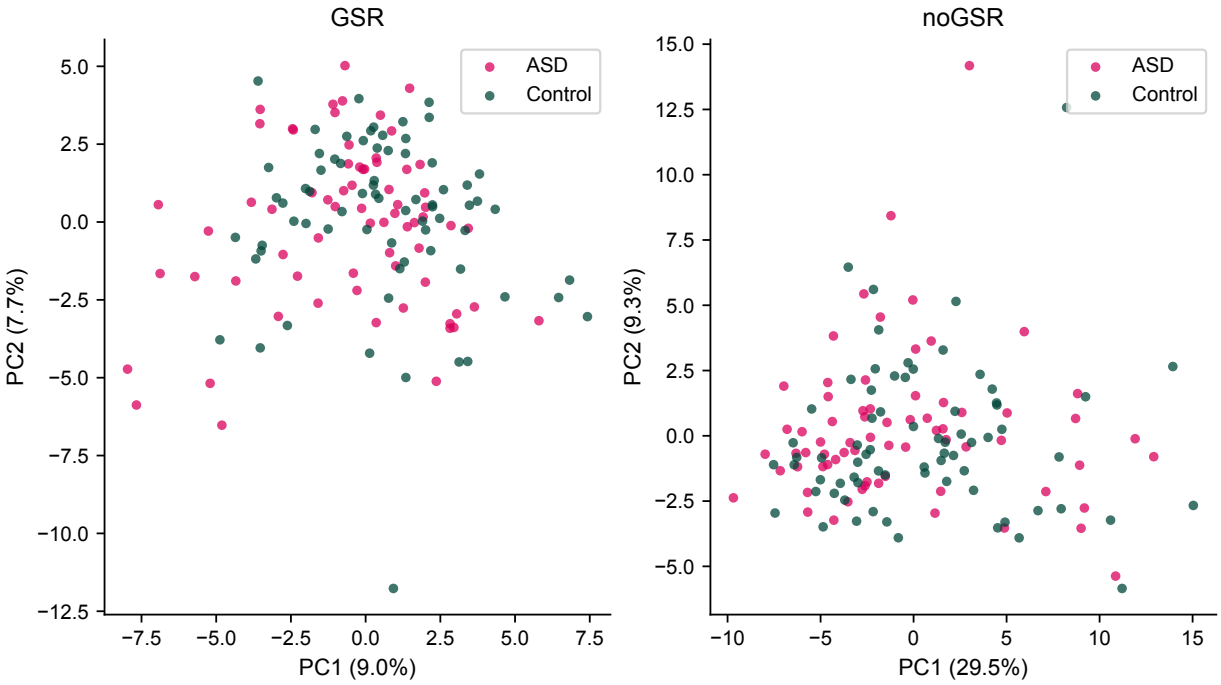


Figure S7: **Descriptive principal-component projection.** PCA uses the standardized raw common 88-feature matrix and all 140 participants separately per pipeline. ASD is magenta and controls dark green. This full-sample descriptive projection is not used in cross-validation, feature selection or inferential testing. Projection coordinates are in S2 Data.

Figure S8 presents exploratory feature-selection performance comparison.

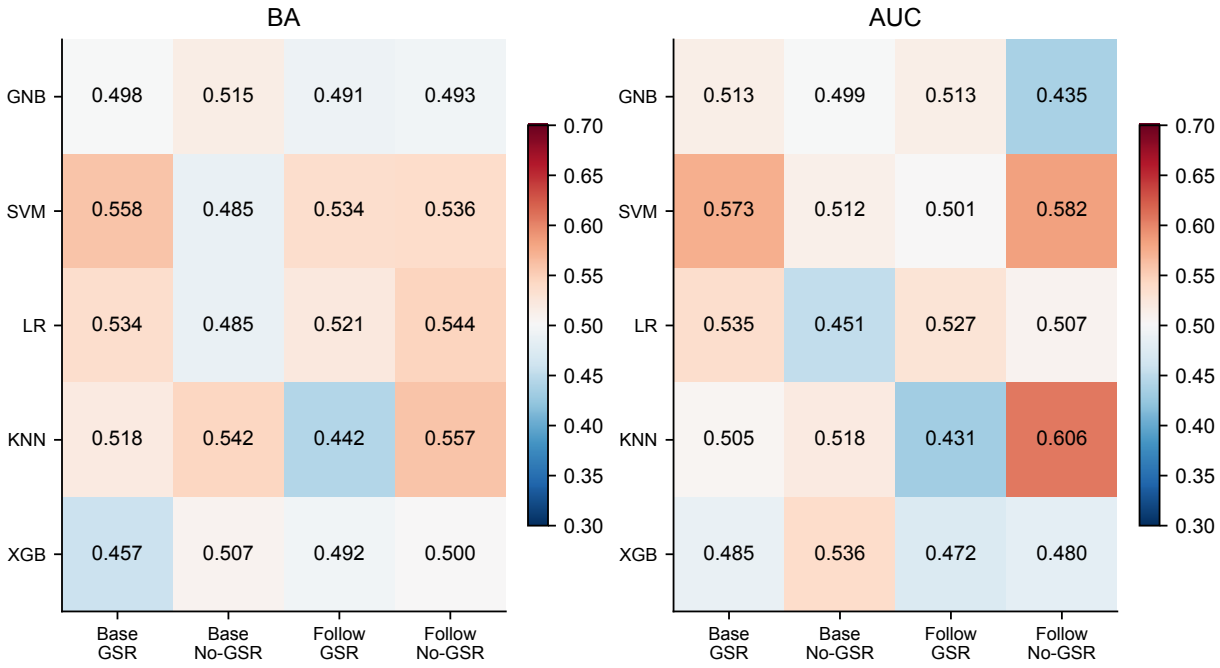


Figure S8: **Exploratory feature-selection performance comparison.** Balanced accuracy and ROC AUC are shown for every baseline and follow-up model/pipeline pair. The follow-up was specified after baseline outer performance was known and is exploratory.

Figure S9 presents exploratory feature-selection roc curves.

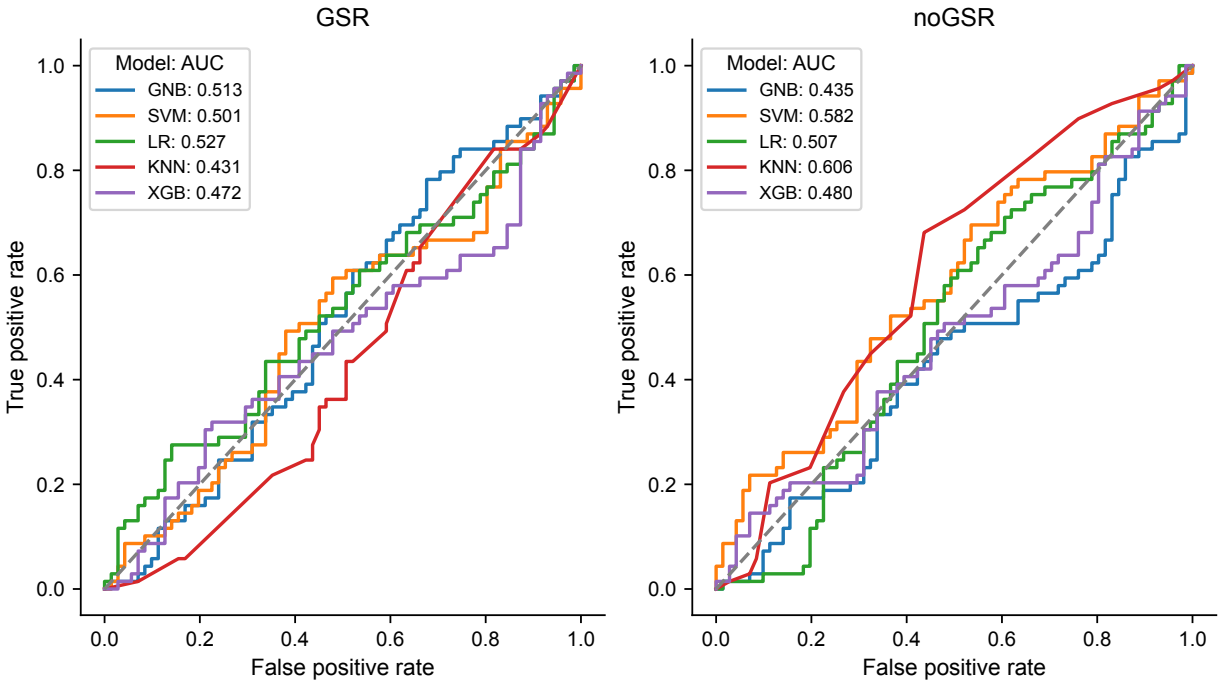


Figure S9: **Exploratory feature-selection ROC curves.** All models use complete out-of-fold scores and the same known outer site folds. A better point estimate does not constitute independent confirmation of improvement.

Figure S10 presents exploratory feature-selection confusion matrices.

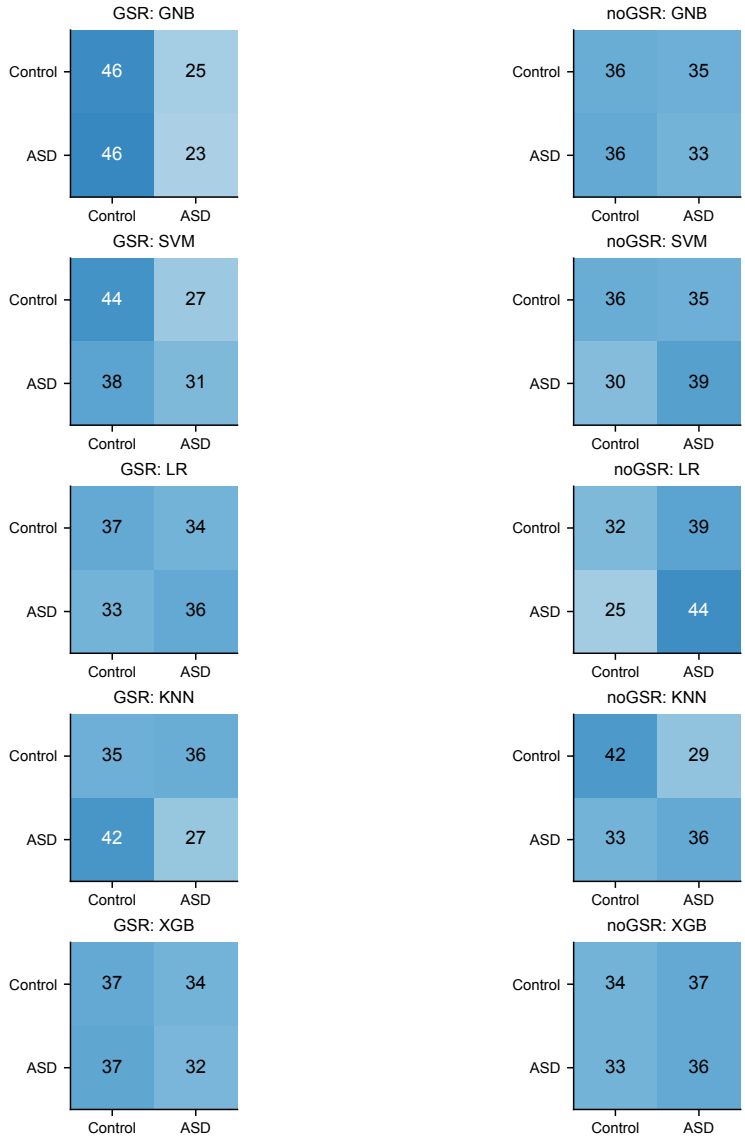


Figure S10: **Exploratory feature-selection confusion matrices.** Counts are printed and color encodes row-normalized proportions, using all 140 out-of-fold predictions per model/pipeline pair. Selection and fitting occur inside training folds.

Figure S11 presents baseline feature-selection frequencies.

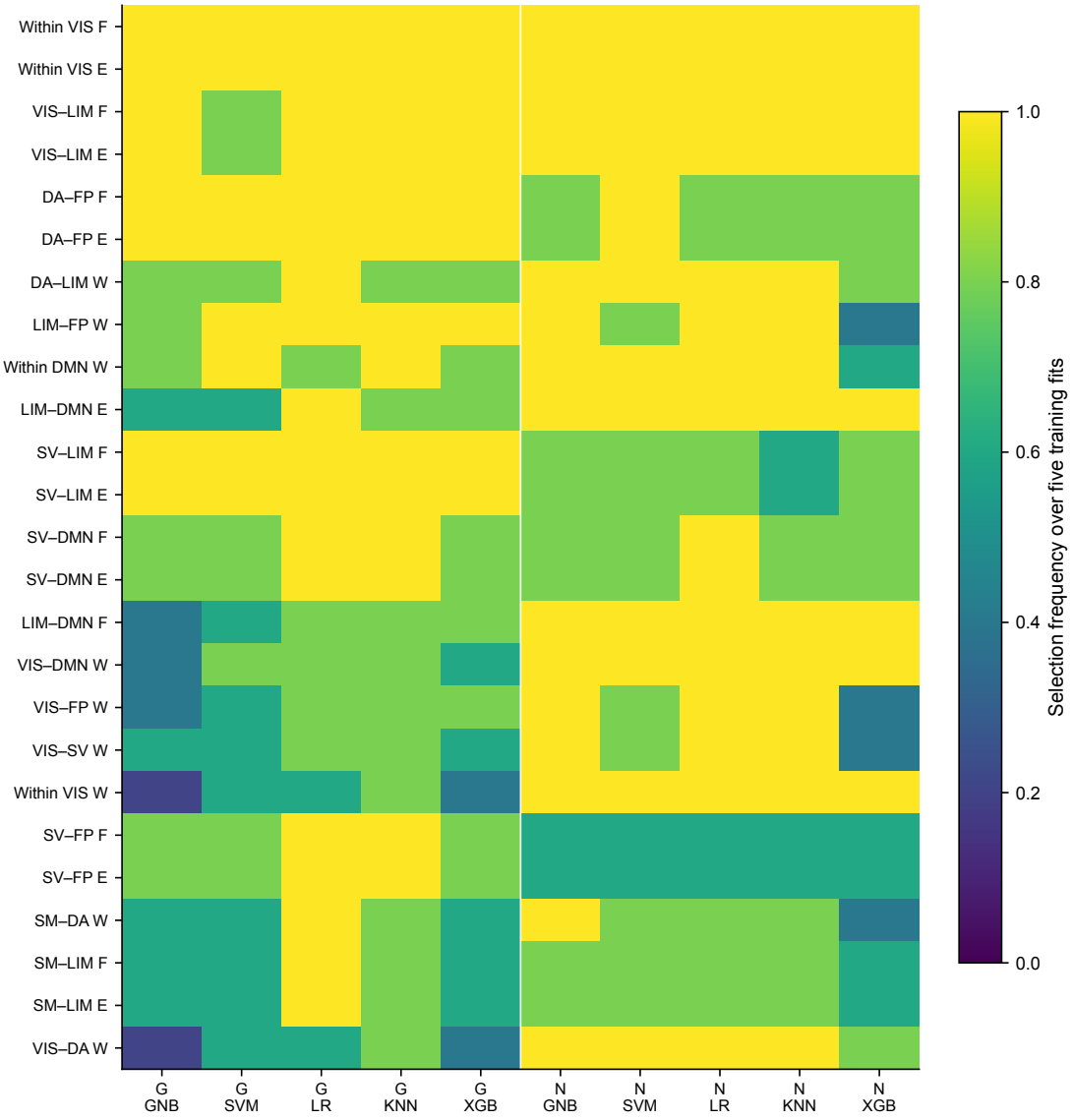


Figure S11: **Baseline feature-selection frequencies.** The 25 most frequently retained features per pipeline are shown across five overlapping outer training sets for each model. Frequencies describe selection, not significance, coefficient magnitude or causal relevance. Full selected names are in S2 Data.

Figure S12 presents follow-up feature-selection frequencies.

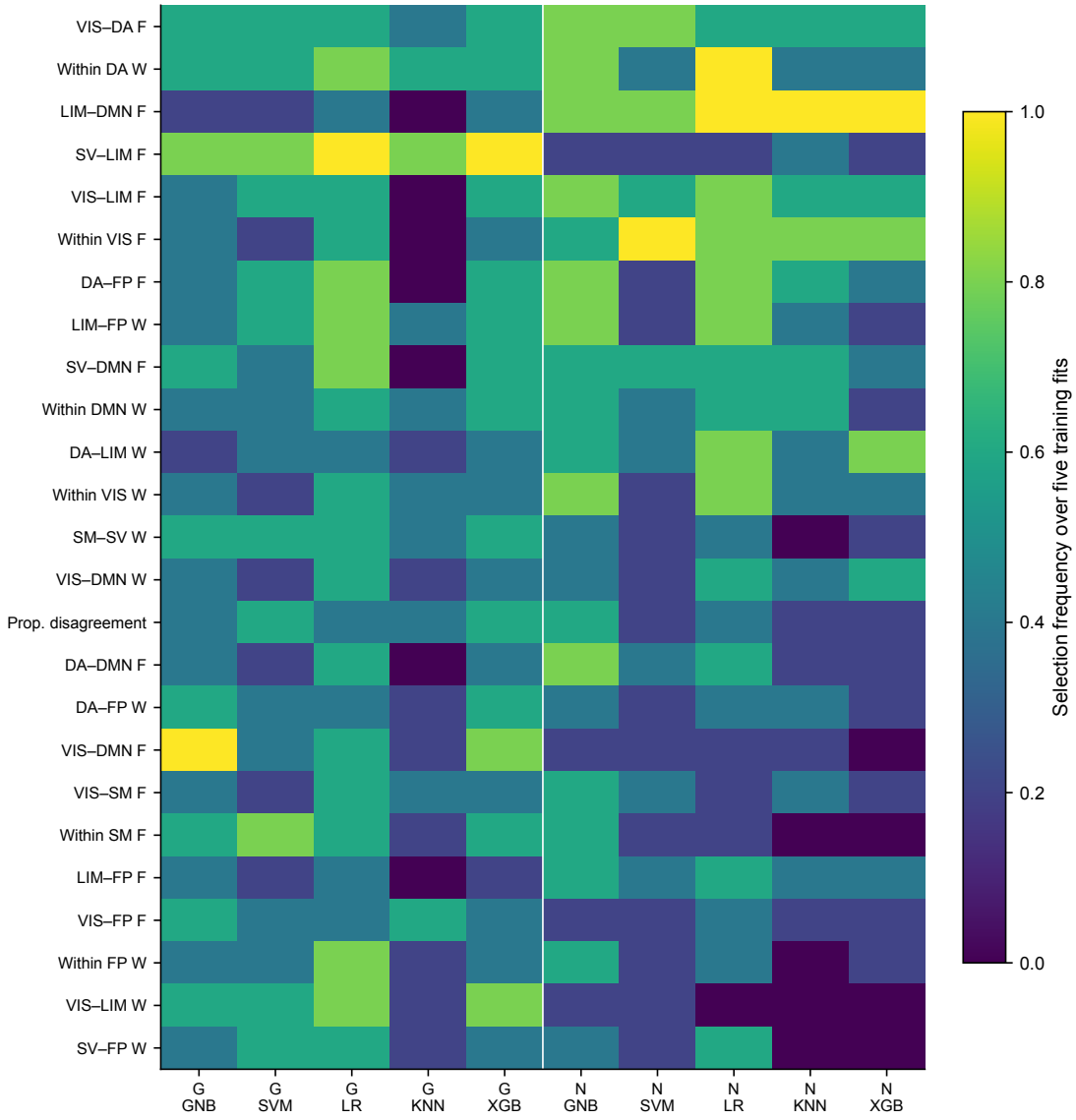


Figure S12: **Follow-up feature-selection frequencies.** The top 25 features are ranked by total retained frequency across model fits. Training-only redundancy pruning precedes the chosen selector. Overlapping folds and multiple procedures preclude interpreting frequencies as independent biomarker evidence.

Figure S13 presents plsc latent covariance summaries.

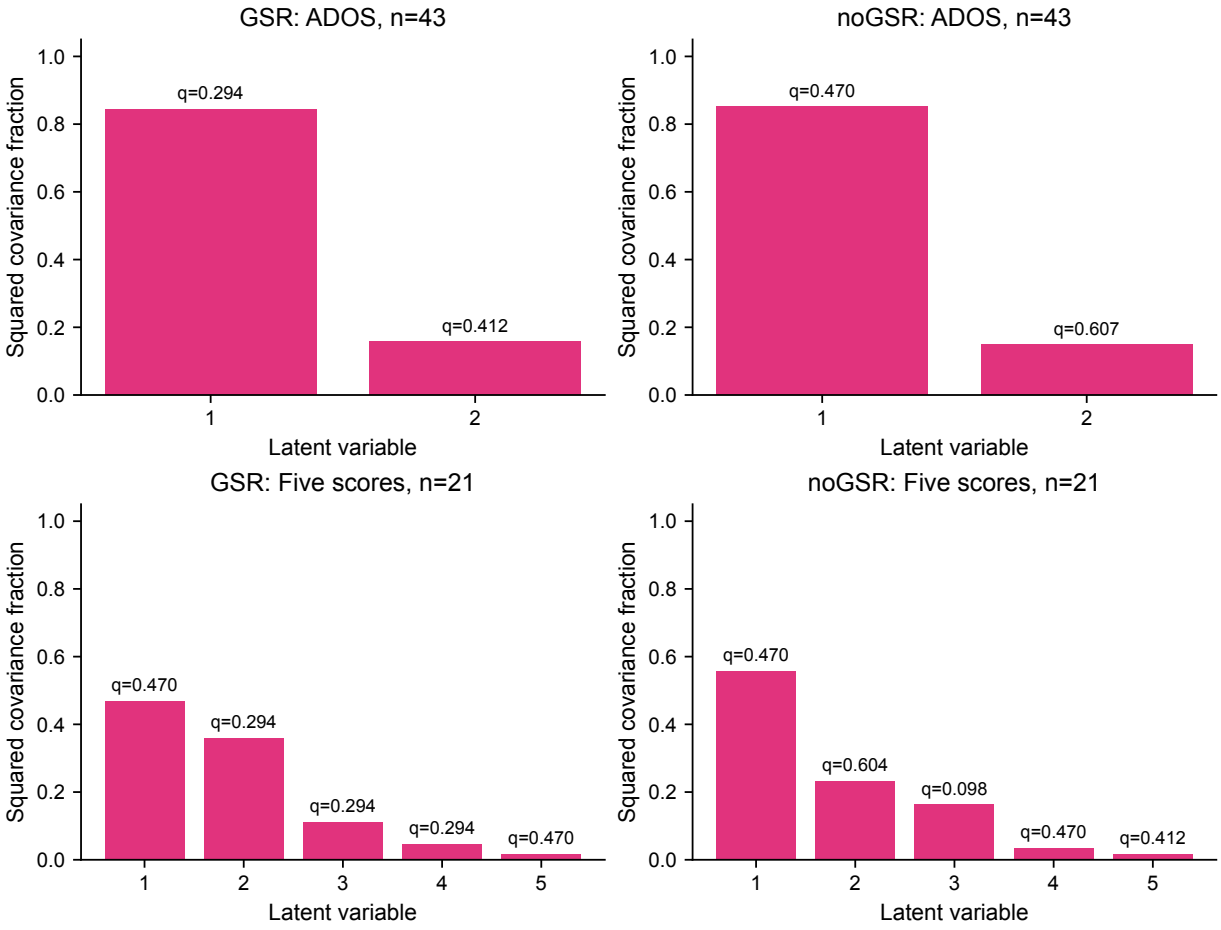


Figure S13: **PLSC latent covariance summaries.** Squared singular-value fractions and corrected rank-wise permutation probabilities are shown for ADOS and five-score complete-case blocks. Sample sizes are 43 and 21. No latent-variable test survives BH across the 14 tests; covariance fractions are not out-of-sample predictive accuracy.